The Project, A to Z

A multi-omics analysis of gastric cancer identifies an externally-validated stromal/fibroblast prognostic program, with cautionary tumor-microbiome and Mendelian-randomization assessments

This is the master reference: what the project asks, what it did, what it found, what it does *not* claim, and where every number lives.

How to trust the numbers here. Every quantitative claim below is checked by `analysis/verify_a_to_z.py`, which re-parses the value from its source file under `results/` (or, for cohort composition, the raw GEO series matrix) and confirms the value as written appears in this document — in the Markdown, Word and PDF renderings simultaneously. The script prints the number of checks and exits non-zero on any mismatch, so you can re-run it yourself rather than take this paragraph on trust. Where a number is quoted, the file it came from is named.

Four companion documents go deeper on specific angles and are not repeated here:

Document	What it is for
`LEARN_MY_PROJECT.md`	Teaching from first principles — DNA, genes, expression, cancer biology
`PROJECT_END_TO_END.md`	Study design and measurement: why paired samples, what pairing does and does not fix
`PROJECT_WALKTHROUGH.md`	Statistical reasoning per analysis, with the traps named
`RESULTS_COMPENDIUM.md`	Every result with a plain-language explanation

PART 1 — THE QUESTION

1.1 What the project asks

Two questions, one primary and one secondary.

Primary (transcriptomic). In gastric cancer, which genes are expressed differently in tumor versus normal stomach; do those genes organize into coherent programs; and does any program carry information about how long a patient survives that is not already contained in standard clinical staging?

Secondary (microbial). Is the community of bacteria living in gastric tissue different in tumors, and if so, is there evidence that any specific microbe *causes* gastric cancer rather than merely accompanying it?

1.2 Why the second question needs a different kind of evidence

An association between a microbe and a tumor has at least four explanations:

1. the microbe causes the cancer;
2. the cancer changes the local environment so the microbe thrives (reverse causation);
3. something else — diet, smoking, age — drives both (confounding);
4. the two groups of samples were processed differently and the "difference" is technical (batch effect).

Sequencing tissue can never distinguish these on its own. That is precisely why the project adds Mendelian randomization, which uses inherited genetic variants — fixed at conception, therefore not caused by a tumor that appears decades later — as instruments for microbial exposure. The design is what licenses causal language, and the result (Part 5) is a clean null.

PART 2 — THE DATA

Arm	Dataset	N	Role
Transcriptome (discovery)	TCGA-STAD	412 tumor / 36 adjacent normal	Differential expression, WGCNA, signature training
Transcriptome (normals)	GTEx v10 stomach	407	Adds normal contrast for the integrated analysis
Survival validation	GSE62254 (ACRG)	300 (152 events)	Independent survival cohort
Survival validation	GSE15459	191 (95 events)	Independent survival cohort
Survival validation	GSE84437	431 (207 events)	Independent survival cohort
DE replication	GSE27342, GSE63089	—	Independent tumor-vs-normal replication
Single cell	GSE134520	43,992 cells	Which cell type expresses the module
Tissue microbiome	DDBJ PRJDB20660	944 samples	16S discovery, raw FASTQ reprocessed
Microbiome validation	PRJNA641258, PRJNA413125	—	Independent 16S cohorts (Italy, Portugal)
GWAS (MR)	IEU OpenGWAS	6 exposures, 1,029 cases	Causal inference on microbial exposures

Survival-cohort sample sizes and event counts above are those used in the analyses and match `results/module_preservation/module_eigengene_cox_external.csv` and `results/meta_HK/meta_inputs.csv`.

One design fact worth stating plainly. The transcriptome cohort and the microbiome cohort share **zero patients** — 415 TCGA patients and 944 microbiome samples with no overlapping identifiers. This is why the project is *multi-omics* in the sense of two parallel arms answering different questions, and why genuine joint factor analysis (MOFA) was assessed and correctly declined: with no shared samples there is nothing to factorize jointly.

PART 3 — THE TRANSCRIPTOMIC CORE

3.1 Differential expression

Of **21,446** genes tested in TCGA-STAD, **2,134 are up-regulated** and **2,362 down-regulated** in tumor versus adjacent normal at $|\log_2 FC| > 1$ and BH-adjusted $p < 0.05$ (`results/tables/TCGA_DEG_results.csv`).

The up-regulated set is dominated by proliferation machinery (MKI67, TPX2, ECT2, KIF14, CENPF, TOP2A) and stromal/matrix genes (COL1A1). The down-regulated set is differentiated gastric tissue: parietal-cell and metabolic genes, AQP4, ADH7, KRT24, CLEC3B, MYOC. Biologically this is the expected signature of a tissue losing its specialized identity and acquiring proliferative, matrix-rich character.

Why 21,446 and not ~20,000. The raw matrix carries 60,660 GENCODE features. After filtering, 21,446 remain, of which 16,164 (75.4%) are protein-coding; the rest are long non-coding RNAs, pseudogenes and unmapped identifiers. The count is not an error — it simply is not a count of protein-coding genes.

Batch sensitivity was tested, not assumed. Adding tissue source site as a covariate changes the DEG list from 4,456 to 4,569 genes with 4,129 shared (92.7%), and log-fold changes correlate at $r = 0.974$. The result does not depend on the batch model.

3.2 The stromal co-expression module (the central finding)

WGCNA on the 5,000 most variable genes at soft power 3 (scale-free $R^2 = 0.88$) detected ten modules. The **red** module is a cancer-associated-fibroblast / extracellular-matrix program, and it is the project's most robust result for one reason: **it replicates externally**.

Module preservation (`modulePreservation`, 200 permutations) gives Zsummary = **15.9** (ACRG), **16.8** (GSE15459), **17.1** (GSE84437) — all far above the $Z > 10$ "strong preservation" threshold. Its eigengene is prognostic in all three independent cohorts, always in the same direction (more stromal activation → worse survival):

Cohort	n (events)	HR per SD	95% CI	p
ACRG/GSE62254	300 (152)	1.274	1.091–1.487	0.00216
GSE15459	191 (95)	1.548	1.248–1.919	6.8×10^{-5}
GSE84437	431 (207)	1.237	1.080–1.416	0.00211

Source: `results/module_preservation/module_eigengene_cox_external.csv`.

The honest caveat, stated in the paper. Within TCGA the module eigengene is prognostic unadjusted (HR/SD 1.307, 95% CI 1.116–1.532, $p = 0.00093$) but **loses significance after**

adjusting for stage and age (HR 1.347, 95% CI 0.954–1.901, p = 0.090; `results/wgcna_real/ME_survival_cox_adjusted.csv`). The program tracks tumor stage and is not cleanly separable from it. The finding is that the program is real, reproducible and prognostic — not that it is independent of staging.

It does not depend on an arbitrary parameter choice. Rebuilding the network at powers 3, 6, 9 and 12 keeps the module survival-associated throughout (HR/SD 1.283–1.308, all p < 0.0024) with 11–12 of 12 hub genes staying inside it.

3.3 Which cell expresses it

Single-cell data answers the "which cell" question that bulk expression cannot. Across 43,992 cells in eight annotated types (`results/scrna/celltype_composition.csv`), **all 23 of 23** hub genes in the non-circular test are fibroblast-dominant, with the dominant-type fraction ranging 0.527–0.998 (median 0.96; `results/scrna/gene_dominant_celltype_noncircular.csv`).

Why "non-circular" matters. If you select genes *because* they are stromal and then show they are stromal, you have proven nothing. The non-circular test removes the genes whose selection depended on stromal identity and asks whether the remainder still localize to fibroblasts. They do. This is the difference between a real result and a tautology, and it is worth understanding as a general lesson.

PART 4 — THE PROGNOSTIC SIGNATURE, REPORTED HONESTLY

4.1 The number that matters

A 25-gene LASSO–Cox signature separates survival in TCGA with an **apparent C-index of 0.72**. That figure is optimistic and the paper says so, because gene screening and LASSO ran once on the whole cohort — the model saw the data it was then scored on.

Under **20 repeats of 5-fold nested cross-validation**, with standardization, screening and tuning rebuilt inside every training fold and performance taken only from untouched out-of-fold predictions, the honest discrimination is:

Metric	Value
Harrell C (ensemble)	**0.6112**
Uno C (ensemble)	0.5727
Harrell C (per-repeat)	0.5981
Integrated Brier score	0.1789

Source: `results/nested_cv/performance.csv`.

The optimism is ~0.11 of C-index. Reporting 0.72 would not be fabrication — it is a real quantity — but presenting it as generalizable performance would be misleading, and that gap is the single most transferable statistical lesson in the project.

4.2 External validation: two of three

Cohort	n (events)	C-index	Log-rank p	HR high-vs-low (95% CI)
TCGA (training)	383 (156)	0.719	2.05×10^{-12}	—
ACRG/GSE62254	300 (152)	0.608	8.6×10^{-5}	1.90 (1.37–2.62)
GSE15459	191 (95)	0.575	0.014	1.68 (1.11–2.54)
GSE84437	431 (207)	0.530	0.46	1.11 (0.84–1.46)

Sources: `results/validation/cindex_comparison.csv`, `results/validation_multi/cindex_HR_summary.csv`.

GSE84437 is a genuine negative, and the project treats it as one. The proposed explanation is range restriction: GSE84437 is 88.7% pT3–T4 disease (384 of 433 staged samples: T1 11, T2 38, T3 92, T4 292; `data/geo/GSE84437_series_matrix.txt.gz`), and a stromal signature discriminates by capturing *variation* in desmoplastic content, which increases with invasion depth. That hypothesis was then **tested directly** by stratifying on pT stage — and it failed: C-index stayed below 0.5 in every stratum. The paper reports this as a tested negative rather than an explained-away artifact. Testing your own escape hatch and reporting that it failed is exactly the behavior reviewers look for.

The pooled effect is not significant. A per-1-SD, age/stage-adjusted random-effects meta-analysis (REML with Hartung–Knapp) across the three external cohorts gives a pooled **HR 1.188 (95% CI 0.962–1.466, p = 0.073)**, I^2 = 19.2% (`results/meta_HK/meta_result.csv`). Directionally consistent, modest, and short of significance under a conservative model.

4.3 Time-varying and predictive structure

In ACRG the signature is prognostic after adjustment for stage and age (HR 1.76, $p = 7.4\times10^{-4}$), but the proportional-hazards assumption is **violated** (cox.zph p = 0.0018; `results/timevarying_ACRG/coxzph.csv`). A time-varying model shows the effect is early (HR/SD 1.49 at 12 months) and attenuates to null later (1.03 at 36 months, 0.87 at 60). It marks earlier mortality, not a durable gradient — a distinction most papers with this data pattern silently skip.

The hazard is also modified by stage (risk × stage LRT p = 0.044), concentrated in advanced and mesenchymal disease; the risk × molecular-subtype interaction is *not* significant (p = 0.094; `results/predictive/interaction_tests.csv`).

4.4 The benchmark against a published signature

A previously published 5-gene gastric-cancer signature, re-tested on identical data under the same leakage-controlled pipeline, scores apparent C = 0.5446 and optimism-corrected **C = 0.5193 —** chance level, with the risk direction inverted (HR high-vs-low 0.627).

One subtlety the project documents rather than hides. The bootstrap optimism correction is implementation-sensitive: the same script and seed give 0.4807 under R 4.3.3 with `rms` 6.8-1, and 0.5193 under R 4.5.3 with `rms` 8.1-1. Both are at chance (0.50 is the no-information value) and the apparent C is identical in both, so the conclusion is unchanged — but both environments are archived (`results/base_paper_replication/sessionInfo.txt` for R 4.5.3 / `rms` 8.1-1 and `sessionInfo_rms6.8-1_R4.3.3.txt` for R 4.3.3 / `rms` 6.8-1) so the discrepancy is inspectable rather than mysterious.

PART 5 — THE MICROBIAL ARM (CAUTIONARY, AND THAT IS THE POINT)

5.1 What the 16S data shows

Differential abundance on CLR-transformed counts finds **44 of 61** genera differing between control and cancer-adjacent mucosa, and **18 of 61** in the paired cancer-adjacent versus tumor comparison (`results/microbiome_biomarker/04a_*.csv`, `04b_*.csv`).

Alpha diversity declines from control to cancer-adjacent mucosa, with the effect size reported rather than only a p-value:

Metric	Median control	Median cancer-adjacent	Cliff's δ	Wilcoxon p
Observed richness	47	30	−0.27	3.3×10^{-7}
Shannon	2.176	1.923	−0.122	0.021
Simpson	0.781	0.756	−0.072	0.172

Note the internal discordance, which the paper discloses: **Simpson diversity does not significantly decline**. Richness falls; evenness-weighted diversity does not clearly follow. Reporting the metric that disagrees is not a weakness — it is what stops a reviewer from finding it for you.

5.2 The batch problem, diagnosed rather than ignored

A random-forest classifier separates cancer from control at **AUC 0.916** (95% CI 0.896–0.936). The same approach predicts **sequencing flowcell** at 77.6% accuracy against a 54.5% majority baseline (`results/microbiome_biomarker/05_rf_metrics_and_batch_sanity.csv`).

That second number is the honest finding. Tumor samples were sequenced on separate flowcells, so a classifier that appears to detect cancer may be detecting the run. Compounding it, tumor tissue is the **richest** group in this dataset — an inversion of the expected biology that is itself evidence of a technical artifact rather than a discovery. This is why the arm is framed as *cautionary*: the analysis is sound, and what it establishes is that the apparent signal cannot be cleanly separated from batch.

5.3 Mendelian randomization: a clean null

Two-sample MR of six microbial exposures against ancestry-matched European gastric cancer (1,029 cases) finds **no significant causal effect anywhere**:

Exposure	nSNP	IVW OR (95% CI)	p
Anti-*H. pylori* IgG seropositivity	17	0.96 (0.71–1.30)	0.79
Streptococcus (genus)	15	1.10 (0.90–1.34)	0.35
Fusobacterium A	23	1.04 (0.79–1.36)	0.79
Prevotella 9	15	0.98 (0.84–1.14)	0.76
Veillonella	8	1.04 (0.84–1.29)	0.69
Lactobacillus	10	0.96 (0.84–1.09)	0.51

Source: `results/mr_real/MR_per_exposure_instruments_REAL.csv`.

Read this as "not established at this power," not "disproved." No exposure reached three genome-wide-significant SNPs, so all six use a locus-wide-suggestive threshold of $p < 1\times10^{-5}$ — a choice that biases toward the null through weak-instrument effects, although the per-SNP F-statistics (minimum 19.3) show the retained instruments are not themselves weak. With 1,029 cases the study is underpowered for modest causal effects. A null under acknowledged low power is a different claim from a null under adequate power, and conflating them is a common error.

Sensitivity diagnostics ran properly: MR-Egger intercepts, Cochran's Q, weighted-median and mode estimators, Steiger filtering, leave-one-out, and MR-PRESSO global tests (all six global p > 0.05, lowest 0.052 for *H. pylori*).

PART 6 — IMMUNE AND DRUG-REPURPOSING ARMS

6.1 Deconvolution validated against pathology

Rather than trusting a deconvolution algorithm, its output was checked against pathologist-scored histology (`results/immune/validation_vs_measured.csv`):

Estimate	Measured against	n	Spearman ρ	p
T cells (MCP)	Leukocyte %	272	0.666	3.6×10^{-36}
ImmuneScore (xCell)	Leukocyte %	272	0.652	2.2×10^{-34}
CD8 T cells (MCP)	Leukocyte %	272	0.468	3.1×10^{-16}
CD8 T cells (MCP)	Lymphocyte infiltration %	274	0.020	0.744

Validation is measure-specific, and that is the finding. The estimates track overall immune burden well and lymphocyte *subset composition* not at all. So the deconvolution should be read as capturing how much immune infiltrate is present, not which subsets compose it. Tumors are enriched for the macrophage/monocyte compartment with no net $CD8^+$ gain, and the CD8 score is **not prognostic** here (Cox HR 1.042, 95% CI 0.944–1.150, p = 0.411; `results/immune/CD8_survival_summary.csv`) — reported as observed rather than dropped for being null.

6.2 Drug repurposing (hypothesis-generating only)

Signature-reversal enrichment against LINCS/CMap and DSigDB nominates CDK4/6, PI3K/mTOR and FGFR inhibitors — palbociclib, NVP-BEZ235, dovitinib, PD-173074 — with orthogonal support from DepMap dependency scores (PIK3CA −0.742, MTOR −1.184, CDK4 −0.825, CDK6 −0.548; `results/depmap/gastric_dependency.csv`, 35 gastric lines). This arm is explicitly hypothesis-generating: no compound was tested in a cell line or animal in this project, and the paper claims nothing beyond a ranked candidate list.

PART 7 — WHAT THE PROJECT DOES NOT CLAIM

Stating this precisely is what makes the rest credible.

1. **Not a novel biological discovery.** The stroma/CAF–prognosis link is well established. The contribution is external replication and honest quantification.
2. **Not a clinically usable test.** Honest C ≈ 0.61 is modest, the pooled external effect is non-significant, and the module is not separable from stage.
3. **No causal microbial claim.** MR is null and underpowered.
4. **No demonstrated therapeutic activity.** Drug candidates are computational.
5. **Not a joint multi-omics factorization.** The two arms share no patients.

PART 8 — REPRODUCIBILITY

- Every reported number maps to a named script in `analysis/` and an output file under `results/` (see `PIPELINE.md` for the full mapping). Script and commit counts are deliberately not quoted here — they change with every commit, so any number written into this document would be stale by the time you read it. Run `git rev-list --count HEAD` and `ls analysis/ | wc -l` for the current values.
- Every result table and figure is committed alongside the code that produced it.
- **Claim verification is executable.** `analysis/verify_a_to_z.py` re-parses each quantitative claim in this document from its source file and confirms the value appears in the Markdown, Word and PDF renderings simultaneously. Run it from the repository root; it exits non-zero on any mismatch.
- **17 figures** (8 main + 9 supplementary), each verified for edge clipping, aspect ratio and text legibility at print size.
- Pinned package versions and `sessionInfo()` dumps archived, including both R/`rms` environments for the version-sensitive benchmark.
- Known gap, stated: two scripts (`17_external_utility_ACRG.R`, `09_functional_enrichment.R`) depend on packages unavailable in the current environment (`dcurves`, `clusterProfiler`), so their committed figures cannot be re-derived here. Their outputs are committed; independent re-execution is unverified.
- Known gap, stated: the harmonized MR SNP data was never persisted, so Figure 8 and Supplementary S8 panels cannot be re-plotted without a live OpenGWAS token.

Outstanding author actions

1. Author names, ORCIDs, corresponding author and CRediT contributions — marked placeholders in the manuscript.
2. Repository URL and a minted DOI (Zenodo or equivalent) at submission.
3. Optional: five of the 17 embedded figures sit above the reference paper's aspect range (height/width > 1.00) and would need re-layout to match — Figure 4 (`Fig7.png`) at 1.38, Supplementary S2 at 1.14, S8 at 1.13, S6 at 1.07 and S7 at 1.04; measured directly from the embedded PNGs.

PART 9 — THE ONE-PARAGRAPH VERSION

Gastric tumors lose differentiated gastric identity and gain proliferative and matrix-rich programs. One of those programs — a fibroblast/extracellular-matrix module — is preserved in three independent cohorts (Zsummary 15.9–17.1) and prognostic in all three, and single-cell data localizes all 23 of its testable hub genes to fibroblasts. A 25-gene signature built from this biology achieves an honest cross-validated C-index of 0.611 (against an apparent 0.72), validates in two of three external cohorts, and does not reach significance when pooled — while a published competitor scores at chance on identical data. In parallel, gastric tissue microbiota differ between tumor and normal, but a classifier predicts sequencing flowcell at 77.6% and tumor tissue is implausibly the richest group, so the signal cannot be separated from batch; and Mendelian randomization finds no causal effect for any of six exposures at the available power. The project's contribution is a replicated stromal prognostic program, reported with its optimism quantified and its negative results intact.